

Assignment: Telecom User Role Management System

Database Administration and SQL Practical Assignment

Domain

Telecommunication Company User Access Management System

Section A — DDL (Data Definition Language)

Question 1

Create a database named telecom_user_mgmt.

```
6 CREATE DATABASE telecom_user_mgmt;  
7 USE telecom_user_mgmt;
```

Question 2

Create a table named users with the following columns:

Column Name	Data Type	Constraint
user_id	Integer	Primary Key, Auto Increment
username	VARCHAR(50)	Unique, Not Null
email	VARCHAR(100)	Unique
department	VARCHAR(50)	
status	VARCHAR(20)	Default ACTIVE
created_at	TIMESTAMP	Default Current Timestamp

```
10 CREATE TABLE users (  
11     user_id INT PRIMARY KEY AUTO_INCREMENT,  
12     username VARCHAR(50) UNIQUE NOT NULL,  
13     email VARCHAR(100) UNIQUE,  
14     department VARCHAR(50),  
15     status VARCHAR(20) DEFAULT 'ACTIVE',  
16     created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP  
17 );
```

Question 3

Create a table named roles with:

- role_id
- role_name
- description

Requirements:

- role_id must be Primary Key
- role_name must be unique

```

20 • CREATE TABLE roles (
21     role_id INT PRIMARY KEY AUTO_INCREMENT,
22     role_name VARCHAR(50) UNIQUE NOT NULL,
23     description VARCHAR(100)
24 );

```

Question 4

Create a table named permissions to store system permissions.

Include:

- permission_id
- permission_name
- module_name

```

27 • CREATE TABLE permissions (
28     permission_id INT PRIMARY KEY AUTO_INCREMENT,
29     permission_name VARCHAR(50),
30     module_name VARCHAR(50)
31 );

```

Question 5

Create a table named user_roles to assign roles to users.

Requirements:

- Create foreign key relationship with users
- Create foreign key relationship with roles

```

34 • CREATE TABLE user_roles (
35     user_id INT,
36     role_id INT,
37     PRIMARY KEY(user_id, role_id),
38     FOREIGN KEY(user_id) REFERENCES users(user_id),
39     FOREIGN KEY(role_id) REFERENCES roles(role_id)
40 );

```

Question 6

Create a table named audit_logs to store user activities.

Requirements:

- Include action tracking
- Include timestamp logging

- Create foreign key with users table

```

43 • CREATE TABLE audit_logs (
44     log_id INT PRIMARY KEY AUTO_INCREMENT,
45     user_id INT,
46     action VARCHAR(100),
47     log_time TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
48     FOREIGN KEY(user_id) REFERENCES users(user_id)
49 );

```

Section B — DML (Data Manipulation Language)

Question 7

Insert 5 users into the users table.

Departments should include:

- IT
- Billing
- Customer Support
- Network Operations
- Management

```

56 • INSERT INTO users (username, email, department) VALUES
57     ('alice', 'alice@telecom.com', 'IT'),
58     ('bob', 'bob@telecom.com', 'Billing'),
59     ('carol', 'carol@telecom.com', 'Customer Support'),
60     ('dave', 'dave@telecom.com', 'Network Operations'),
61     ('eve', 'eve@telecom.com', 'Management');

```

Question 8

Insert the following roles:

Role

Admin

Billing Officer

Support Agent

Network Engineer

Viewer

```

64 • INSERT INTO roles (role_name, description) VALUES
65 ('Admin', 'System administrator'),
66 ('Billing Officer', 'Handles billing operations'),
67 ('Support Agent', 'Customer support staff'),
68 ('Network Engineer', 'Manages network infrastructure'),
69 ('Viewer', 'Read-only access');

```

Question 9

Insert at least 5 permissions into the permissions table.

Example modules:

- User Management
- Billing
- Network
- Reports

```

72 INSERT INTO permissions (permission_name, module_name) VALUES
73 ('Manage Users', 'User Management'),
74 ('Process Billing', 'Billing'),
75 ('Monitor Network', 'Network'),
76 ('View Reports', 'Reports'),
77 ('Audit Logs Access', 'Audit');

```

Question 10

Assign roles to all users using the user_roles table.

```

80 • INSERT INTO user_roles VALUES
81 (1, 1), -- alice → Admin
82 (2, 2), -- bob → Billing Officer
83 (3, 3), -- carol → Support Agent
84 (4, 4), -- dave → Network Engineer
85 (5, 5); -- eve → Viewer

```

Question 11

Update one user status from ACTIVE to INACTIVE.

```

88 • UPDATE users SET status = 'INACTIVE' WHERE username = 'eve';

```

Question 12

Delete one permission record from the permissions table.

```

91 • DELETE FROM permissions WHERE permission_name = 'Audit Logs Access';

```

Section C — DQL (Data Query Language)

Question 13

Display all users from the users table.

98 • **SELECT * FROM users;**

Result Grid						
		Filter Rows:		Edit:		Export/Import:
						Wrap Cell Content:
	user_id	username	email	department	status	created_at
▶	1	alice	alice@telecom.com	IT	ACTIVE	2026-05-10 23:00:09
	2	bob	bob@telecom.com	Billing	ACTIVE	2026-05-10 23:00:09
	3	carol	carol@telecom.com	Customer Support	ACTIVE	2026-05-10 23:00:09
	4	dave	dave@telecom.com	Network Operations	ACTIVE	2026-05-10 23:00:09
	5	eve	eve@telecom.com	Management	INACTIVE	2026-05-10 23:00:09
*	NULL	NULL	NULL	NULL	NULL	NULL

Question 14

Display only active users.

101 • **SELECT * FROM users WHERE status = 'ACTIVE';**

Result Grid						
		Filter Rows:		Edit:		Export/Import:
						Wrap Cell Content:
	user_id	username	email	department	status	created_at
▶	1	alice	alice@telecom.com	IT	ACTIVE	2026-05-10 23:00:09
	2	bob	bob@telecom.com	Billing	ACTIVE	2026-05-10 23:00:09
	3	carol	carol@telecom.com	Customer Support	ACTIVE	2026-05-10 23:00:09
	4	dave	dave@telecom.com	Network Operations	ACTIVE	2026-05-10 23:00:09
*	NULL	NULL	NULL	NULL	NULL	NULL

Question 15

Display users together with their assigned roles using JOIN.

Expected columns:

- username
- department
- role_name

```
104 • SELECT u.username, u.department, r.role_name
105 FROM users u
106 JOIN user_roles ur ON u.user_id = ur.user_id
107 JOIN roles r ON ur.role_id = r.role_id;
```

Result Grid			
		Filter Rows:	
		Export:	
		Wrap Cell Content:	
	username	department	role_name
▶	alice	IT	Admin
	bob	Billing	Billing Officer
	carol	Customer Support	Support Agent
	dave	Network Operations	Network Engineer
	eve	Management	Viewer

Question 16

Count total users by department.

```

110 • SELECT department, COUNT(*) AS total_users
111     FROM users GROUP BY department;
112

```

Result Grid		
	Filter Rows:	Export: Wrap Cell Content:
	department	total_users
▶	IT	1
	Billing	1
	Customer Support	1
	Network Operations	1
	Management	1

Question 17

Display all users from the Billing department.

```

114 • SELECT * FROM users WHERE department = 'Billing';
115

```

Result Grid						
	Filter Rows:	Edit:	Export/Import:	Wrap Cell Content:		
	user_id	username	email	department	status	created_at
▶	2	bob	bob@telecom.com	Billing	ACTIVE	2026-05-10 23:00:09
*	NULL	NULL	NULL	NULL	NULL	NULL

Question 18

Display all roles and their permissions.

```

117 • SELECT r.role_name, p.permission_name, p.module_name
118     FROM roles r
119     JOIN permissions p ON r.role_id = p.permission_id;

```

Result Grid			
	Filter Rows:	Export:	Wrap Cell Content:
	role_name	permission_name	module_name
▶	Admin	Manage Users	User Management
	Billing Officer	Process Billing	Billing
	Support Agent	Monitor Network	Network
	Network Engineer	View Reports	Reports

Question 19

Sort all users by username in ascending order.

```

122 • SELECT * FROM users ORDER BY username ASC;

```

Result Grid						
	Filter Rows:	Edit:	Export/Import:	Wrap Cell Content:		
	user_id	username	email	department	status	created_at
▶	1	alice	alice@telecom.com	IT	ACTIVE	2026-05-10 23:00:09
	2	bob	bob@telecom.com	Billing	ACTIVE	2026-05-10 23:00:09
	3	carol	carol@telecom.com	Customer Support	ACTIVE	2026-05-10 23:00:09
	4	dave	dave@telecom.com	Network Operations	ACTIVE	2026-05-10 23:00:09
	5	eve	eve@telecom.com	Management	INACTIVE	2026-05-10 23:00:09
*	NULL	NULL	NULL	NULL	NULL	NULL

Question 20

Display the total number of users in the system.

```
125 • SELECT COUNT(*) AS total_users FROM users;
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
total_users			
5			

Section D — DTL / TCL (Transaction Control Language)

Question 21

Start a transaction and insert a record into the audit_logs table.

Commit the transaction.

```
132 • START TRANSACTION;
133 • INSERT INTO audit_logs (user_id, action) VALUES (1, 'Login');
134 • COMMIT;
135
136 • SELECT * FROM audit_logs;
```

Result Grid	Filter Rows:	Edit:	Export/Import:	Wrap Cell Content:
log_id	user_id	action	log_time	
1	1	Login	2026-05-10 23:09:39	
NULL	NULL	NULL	NULL	

Question 22

Start a transaction and update one user status.

Rollback the transaction.

Verify whether the data is restored.

```
139 • START TRANSACTION;
140 • UPDATE users SET status = 'INACTIVE' WHERE username = 'bob';
141 • ROLLBACK;
142 • SELECT * FROM users WHERE username = 'bob';
```

Result Grid

Filter Rows:

Edit:

Export/Import:

Wrap Cell Content:

	user_id	username	email	department	status	created_at
▶	2	bob	bob@telecom.com	Billing	ACTIVE	2026-05-10 23:00:09
*	NULL	NULL	NULL	NULL	NULL	NULL

Question 23

Use SAVEPOINT during multiple update operations.

Requirements:

- Create one savepoint
- Rollback to savepoint
- Commit final transaction

```

145 • START TRANSACTION;
146 • UPDATE users SET status = 'INACTIVE' WHERE username = 'carol';
147 • SAVEPOINT sp1;
148 • UPDATE users SET status = 'INACTIVE' WHERE username = 'dave';
149 • ROLLBACK TO sp1;
150 • COMMIT;
151
152 • SELECT * FROM users;

```

Result Grid Filter Rows: Edit: Export/Import: Wrap Cell Content:						
	user_id	username	email	department	status	created_at
▶	1	alice	alice@telecom.com	IT	ACTIVE	2026-05-10 23:00:09
	2	bob	bob@telecom.com	Billing	ACTIVE	2026-05-10 23:00:09
	3	carol	carol@telecom.com	Customer Support	INACTIVE	2026-05-10 23:00:09
	4	dave	dave@telecom.com	Network Operations	ACTIVE	2026-05-10 23:00:09
	5	eve	eve@telecom.com	Management	INACTIVE	2026-05-10 23:00:09
*	NULL	NULL	NULL	NULL	NULL	NULL

Question 24

Demonstrate the difference between COMMIT and ROLLBACK using practical examples.

```

152 • SELECT * FROM users;
153 -- Q24: COMMIT vs ROLLBACK demo
154 • START TRANSACTION;
155 • UPDATE users SET department = 'IT' WHERE username = 'eve';
156 • COMMIT;
157
158 • START TRANSACTION;
159 • UPDATE users SET department = 'HR' WHERE username = 'bob';
160 • ROLLBACK;

```

Result Grid Filter Rows: Edit: Export/Import: Wrap Cell Content:						
	user_id	username	email	department	status	created_at
▶	1	alice	alice@telecom.com	IT	ACTIVE	2026-05-10 23:00:09
	2	bob	bob@telecom.com	Billing	ACTIVE	2026-05-10 23:00:09
	3	carol	carol@telecom.com	Customer Support	INACTIVE	2026-05-10 23:00:09
	4	dave	dave@telecom.com	Network Operations	ACTIVE	2026-05-10 23:00:09
	5	eve	eve@telecom.com	IT	INACTIVE	2026-05-10 23:00:09
*	NULL	NULL	NULL	NULL	NULL	NULL

Section E — DCL (Data Control Language)

Question 25

Create a database user named `billing_user`.

```
167 • CREATE USER 'billing_user'@'localhost' IDENTIFIED BY 'password123';
```

Question 26

Grant `SELECT` and `UPDATE` privileges on the `users` table to `billing_user`.

```
170 • GRANT SELECT, UPDATE ON telecom_user_mgmt.users TO 'billing_user'@'localhost';
```

Question 27

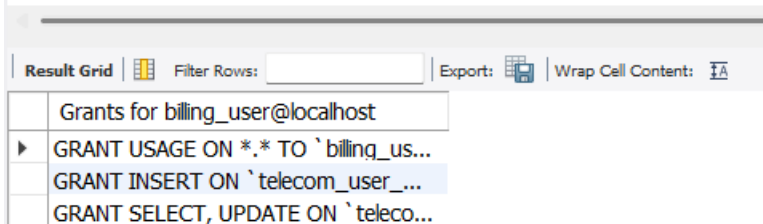
Grant `INSERT` permission on the `audit_logs` table to `billing_user`.

```
173 • GRANT INSERT ON telecom_user_mgmt.audit_logs TO 'billing_user'@'localhost';
```

Question 28

Display all granted privileges for `billing_user`.

```
176 • SHOW GRANTS FOR 'billing_user'@'localhost';
```



Result Grid	Filter Rows:	Export:	Wrap Cell Content:
Grants for billing_user@localhost			
▶	GRANT USAGE ON *.* TO `billing_us...		
	GRANT INSERT ON `telecom_user_...		
	GRANT SELECT, UPDATE ON `teleco...		

Question 29

Revoke `UPDATE` privilege from `billing_user`.

```
179 • REVOKE UPDATE ON telecom_user_mgmt.users FROM 'billing_user'@'localhost';
```

Question 30

Create another database user named `viewer_user` with read-only access to the entire database.

```
182 • CREATE USER 'viewer_user'@'localhost' IDENTIFIED BY 'password123';
```

```
183 • GRANT SELECT ON telecom_user_mgmt.* TO 'viewer_user'@'localhost';
```

Section F — Practical Scenario Questions

Question 31

The telecom company wants only Admin users to manage user accounts.

Write the SQL privilege commands required.

```
190 • CREATE USER 'admin'@'localhost' IDENTIFIED BY 'password123';
```

```
191 • GRANT ALL PRIVILEGES ON telecom_user_mgmt.users TO 'admin'@'localhost';
```

Question 32

The Billing department should only view and update billing-related records.

Write the required `GRANT` statements.

```
194 • GRANT SELECT, UPDATE ON telecom_user_mgmt.users TO 'billing_user'@'localhost';
```

Question 33

Support Agents should only have read access.

Write the SQL commands.

```
197 • CREATE USER 'support_agent'@'localhost' IDENTIFIED BY 'password123';
198 • GRANT SELECT ON telecom_user_mgmt.* TO 'support_agent'@'localhost';
```

Question 34

Generate an SQL query to display:

- Username
- Role Name
- Permission Name
- Module Name

Using multiple JOIN operations.

```
201 • SELECT u.username, r.role_name, p.permission_name, p.module_name
202 FROM users u
203 JOIN user_roles ur ON u.user_id = ur.user_id
204 JOIN roles r ON ur.role_id = r.role_id
205 JOIN permissions p ON r.role_id = p.permission_id;
```

Result Grid				
Filter Rows:		Export:		Wrap Cell Content: IA
	username	role_name	permission_name	module_name
▶	alice	Admin	Manage Users	User Management
	bob	Billing Officer	Process Billing	Billing
	carol	Support Agent	Monitor Network	Network
	dave	Network Engineer	View Reports	Reports

Question 35

Explain the importance of Role-Based Access Control (RBAC) in telecom systems.

Include:

- Security
- Auditability
- Data protection
- Access limitation
- Compliance

Role-Based Access Control (RBAC) ensures:

- Security: restricts unauthorized access
- Auditability: logs user actions
- Data protection: prevents misuse
- Access limitation: least privilege principle
- Compliance: meets regulatory standards